Project Initiation

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Project Initiation

- Short burst of activity at the start of the project to
  - Define the project goals, scope, and constraints
  - Identify the main project's stakeholders
  - Check the project is worthwhile and viable before committing more resources
- Can take a few hours to several days

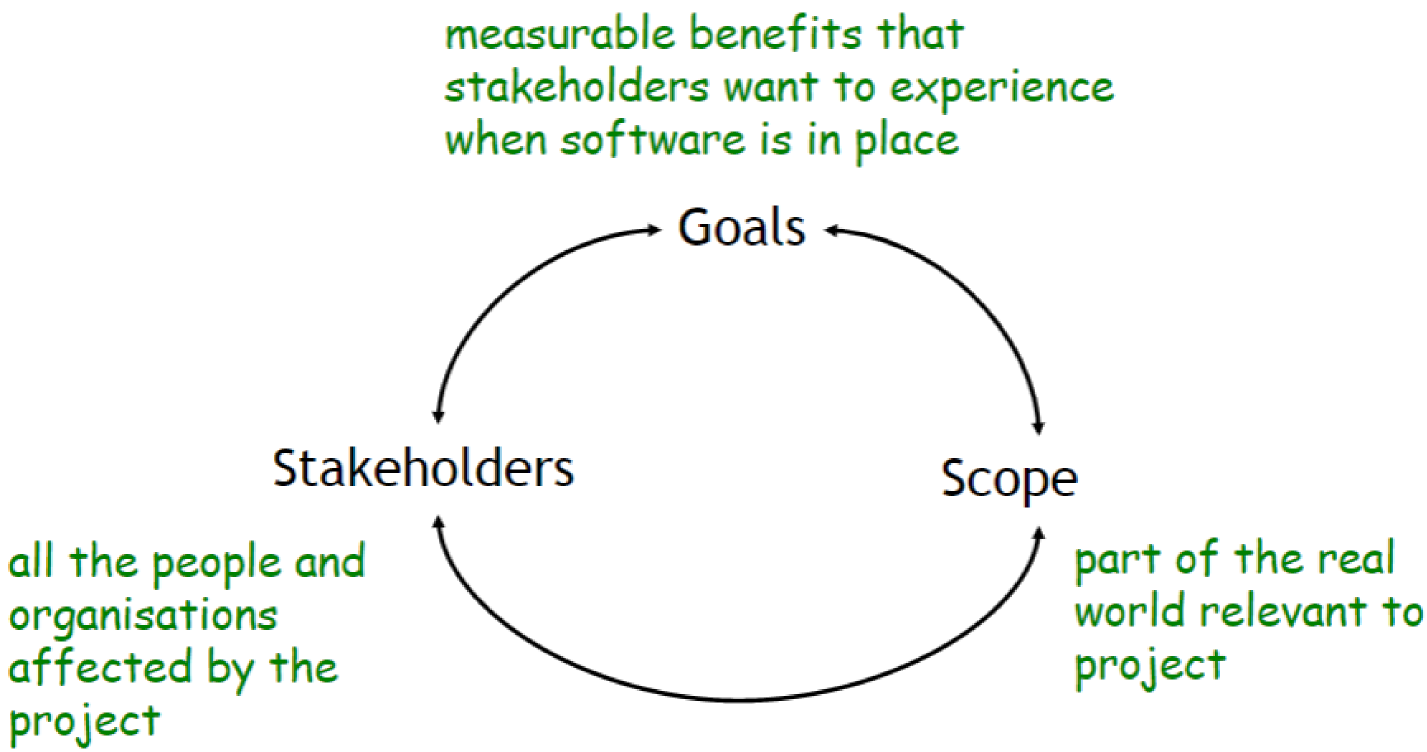
Project Initiation Questions

- What type of project?
- Who are the stakeholders?
- What are the top-level goals?
- What is the project scope?
- What are the main conflicts and major risks?
- What are the budget and schedule timelines?
- Is the project worth doing?

What type of project?

- Greenfield vs. brownfield (new system vs. system evolution)
- Customer-driven vs. market driven
- In-house vs. outsourced
- Single-product vs. product-line
- Problem-driven vs. technology-driven

Goals, scope, and stakeholders are linked



Goals: What do you want to achieve?

PAM – Purpose, Advantage, Measurement

- Purpose
  - What is the product supposed to do?
- Advantages
  - What business advantage should it provide to the client?
  - What advantage should it provide to other stakeholders?
- Measurement
  - How will the advantages be measured?

An Ambulance Despatching System

- Purpose
  - New CAD will handle encoding of urgent calls and assist allocators despatching ambulances to incidents
- Advantage
  - Faster and more accurate allocation of ambulances
  - Faster, more reliable communication with ambulances
- Measurement
  - Government standard: first ambulance must arrive at incident scene within 14 minutes for at least 95% of cases
  - ...

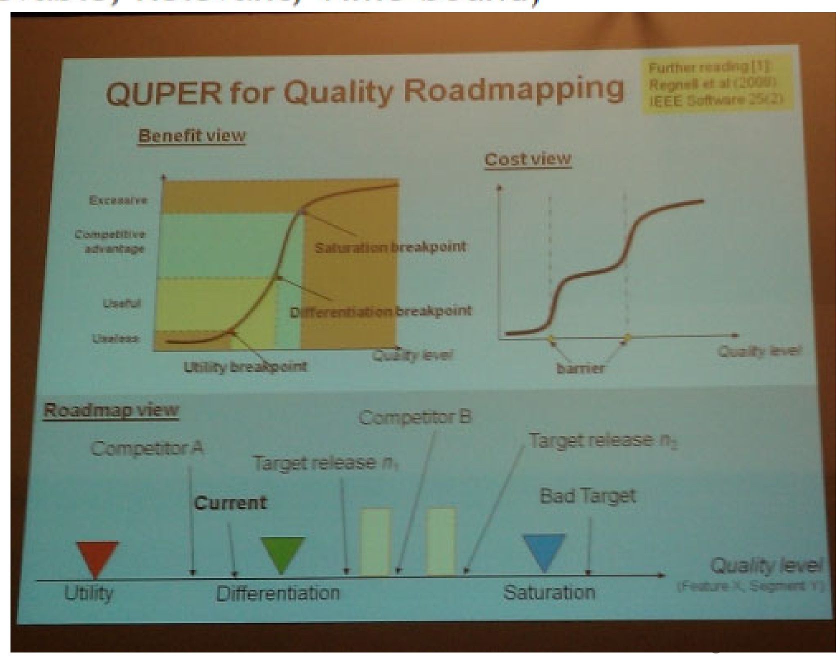
Languages for specifying measurable goals

Informal

- SMART goals (Specific, Measurable, Achievable, Relevant, Time-bound)
- Fit Criteria

Formal

- KAOS objective functions
- QuPer



- Benefit view, Roadmap View, Cost View

Fit Criteria Examples

- A fit criterion for a functional requirement specifies the completion of the function of the product that is specified by the functional requirement.
- Example 1: FR fit criteria

- if the required function is to: send an email to the student after a marked TMA has been uploaded by the tutor, then the

- fit criterion for this requirement is that an email should indeed be sent to the student and reflected in their mailbox on the forum.

Another Fit Criteria Example

- A fit criterion for a non-functional requirement specifies a value or values, on a particular scale of measurement, that must be attained by the property or quality with which the requirement is concerned.
- Example 2: NFR fit criteria

- If NFR is: The credit card number should be accepted securely. Then the
- fit-criteria might be: The credit card number should be revealed to a third party in less than 0.0001% of cases.

KAOS Objective functions

Goal Achieve [Ambulance Intervention]

Desired Behaviour

GIVEN an incident has occurred
WHEN the ambulance service receives a call reporting the incident
THEN within 14 minutes
a first ambulance arrives at the incident scene

Quality Variable

response_time: Incident -> Time

Objective Function

Maximise 14_min_response_rate = *P*(response_time <= 14')

	Current	Must	Target	Ideal
14_min_response rate	82%	82%	95%	99%

Beware of Projects with Unclear Goals

- Common errors

  1. Focus on the product/technology without clarifying the advantages in the application domain

     "The project's objective is to develop a new, modern ambulance despatching system using the latest technologies"

  2. Vague, ambiguous, unmeasurable advantages

     "The project's objective is to improve efficiency"

- Risks: project that lacks vision and direction, wasted effort, building the wrong system, system brings little or no value to business

Practical hint

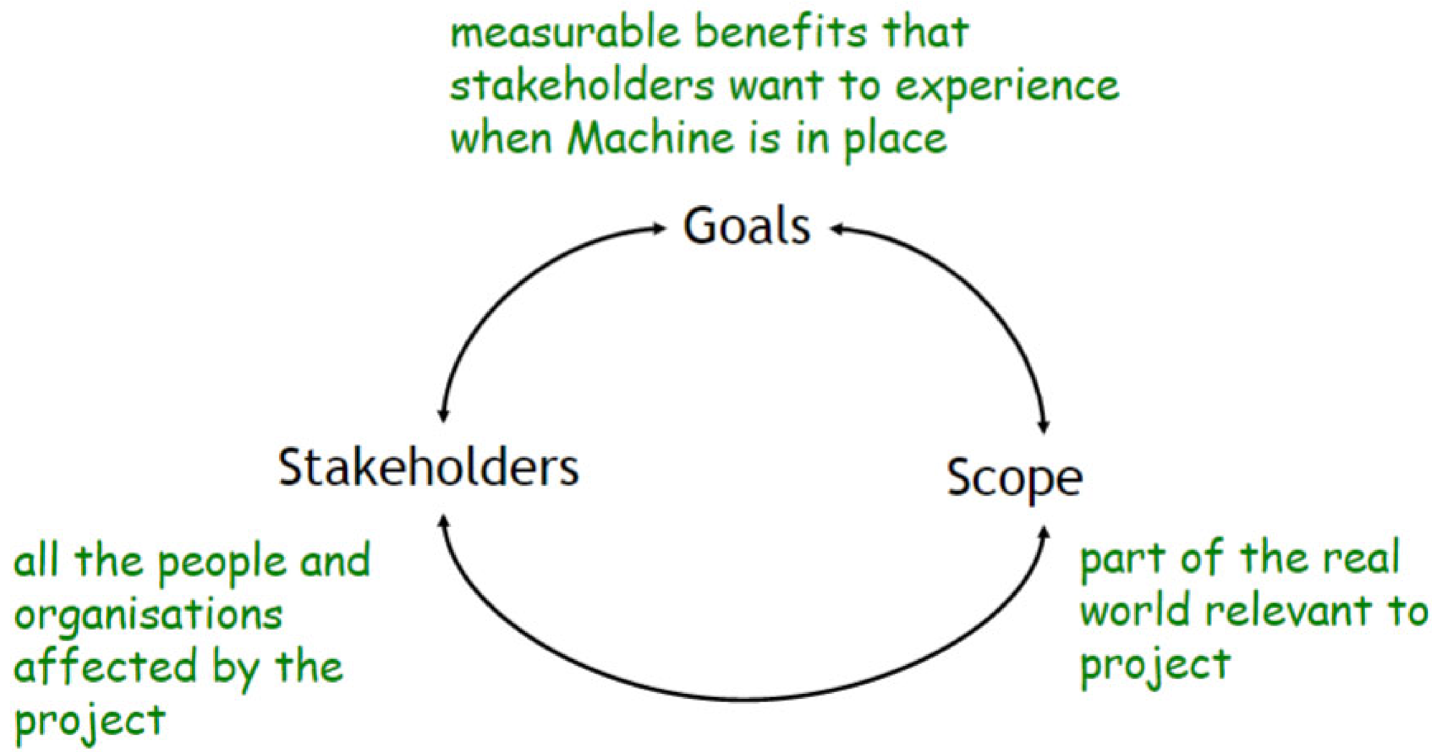
- To define measurable goals, identify what observable changes stakeholders want to see in the world
- Example: if your stakeholders are concerned with a vague goal X, ask the following questions
  - Why do you want more (or less) X
  - What differences would you see between a world that has more X and a world that has less X

Exercise

Identify plausible fit criteria for the following goal

The new CAD system shall help reduce stress among ambulance staff

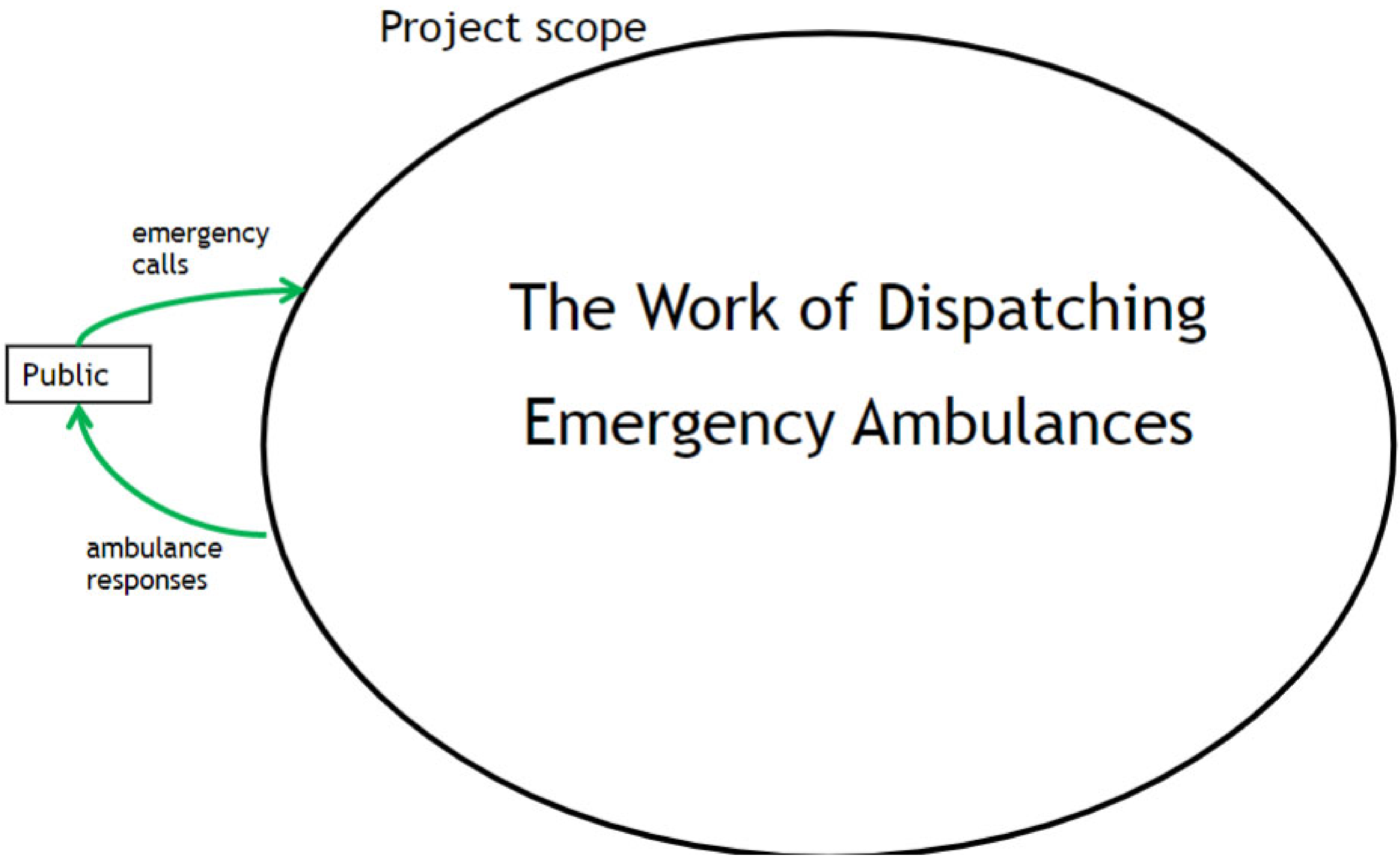
Stakeholders, Goals, Scope



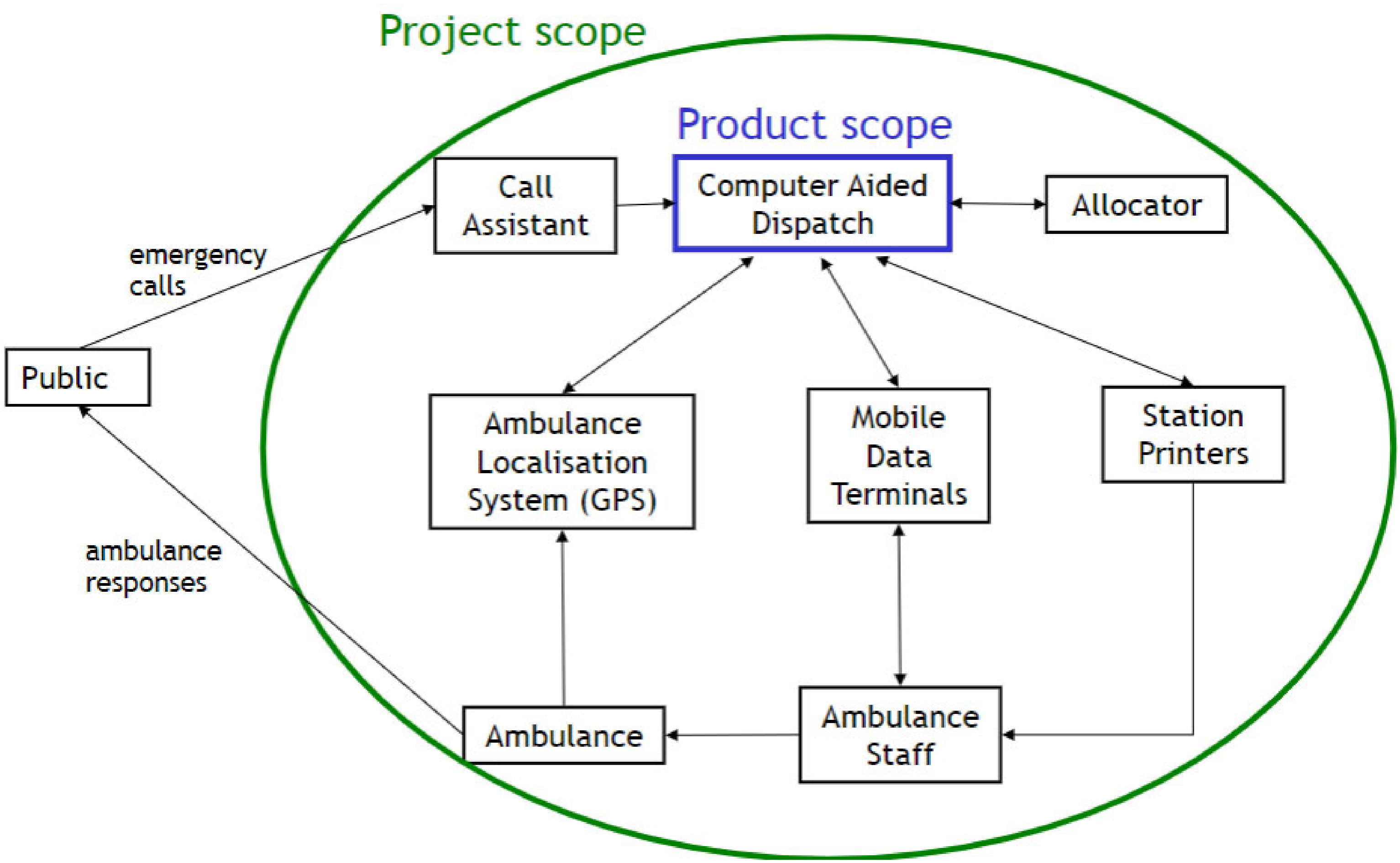
Scope: defining the project's boundary

- the project scope is the part of the real world you need to investigate and that you might change to satisfy the project goals
- the product scope is an outline of the main areas of functionalities and interfaces for the envisioned product

Modelling Scope with Context Diagrams



Modelling Scope with Context Diagrams



Beware of Scoping Errors

- Project scope too narrow implies risk of building the wrong system, prevents exploration of alternatives
  - Usually wrong if product scope = project scope
  - NASA's multimillion dollar zero-gravity pen vs. Russian cosmonauts' pencils (this is a myth but explains the idea
- Project scope too wide implies risk transcending project authority, lack of coherent vision, requirements creep, and wasted effort
  - The ultimate top level goals: "Make everyone happy", "Bring peace on earth", ...
  - The parable of the toaster (Look for it on the web)

http://www.solipsys.co.uk/new/

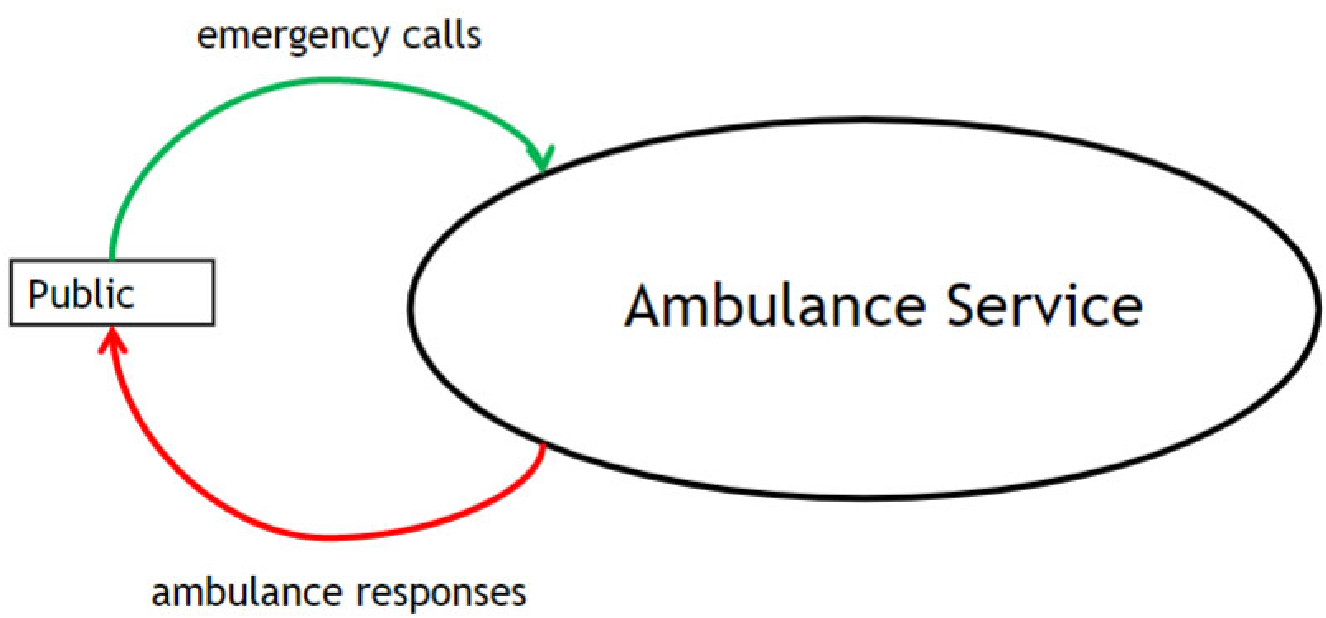
http://www.solipsys.co.uk/new/TheParableOfTheToaster.html

Practical hints

- Useful to start from context diagram of existing system
- Derive the project scope from the project goals
  - the world phenomena used to define goal measurement become inputs and outputs of the project context diagram
- Identify project goals from the project scope
  - from the project context diagram, identify goals that concern the inputs and outputs of this diagram

Example relation between project goal and scope

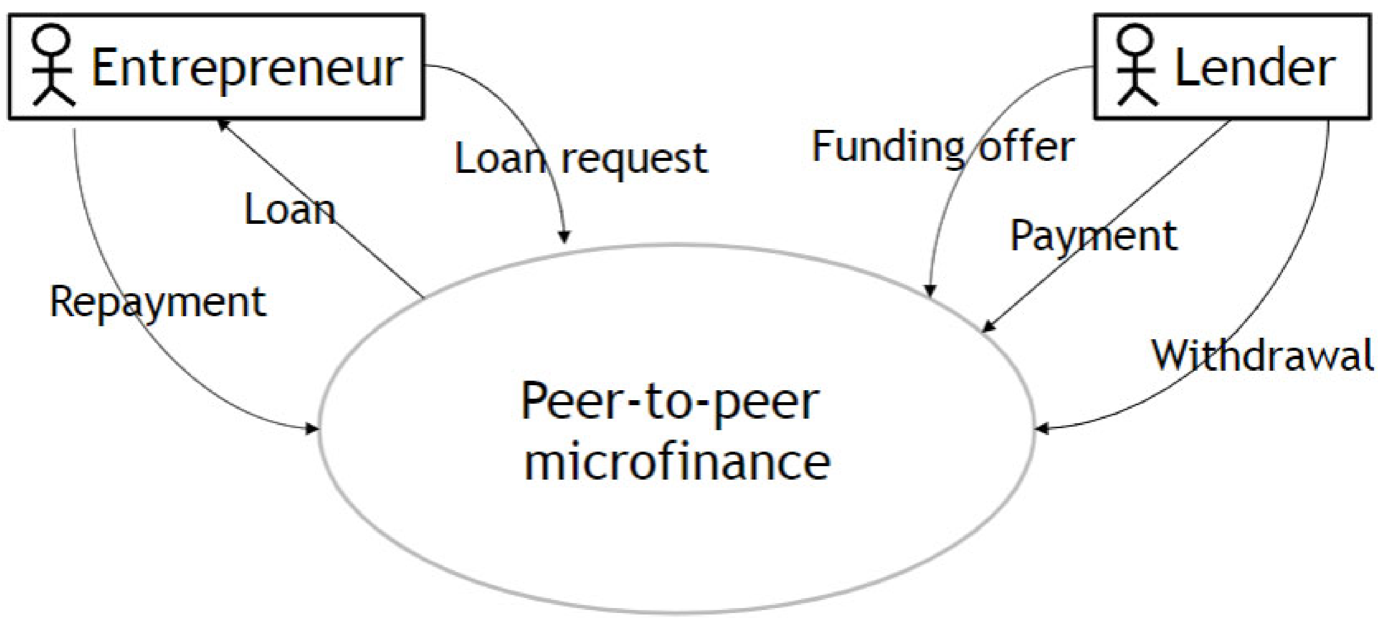
Goal: ambulance must arrive at incident scene within 14 minutes after first emergency call reporting the incident



Exercise: from project scope to goals

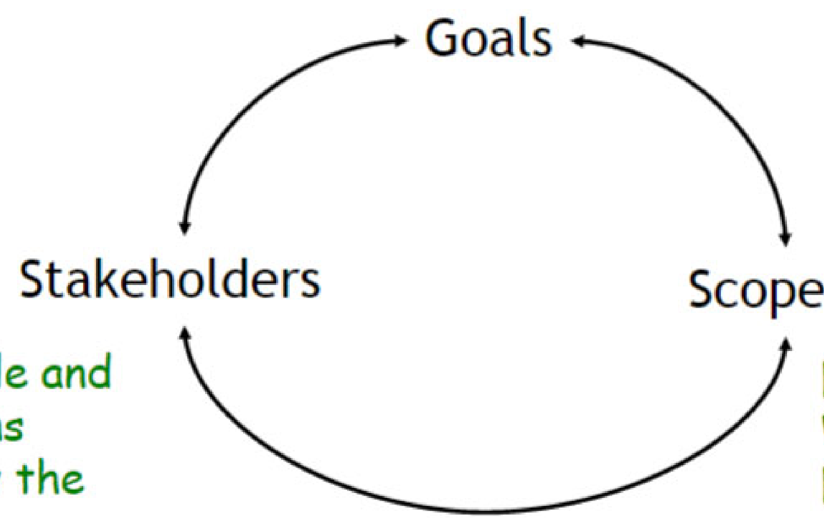
Consider a peer-to-peer microfinance system

Identify plausible goals from with this project-level context diagram



Stakeholders, Goals, Scope

measurable benefits that stakeholders want to experience when Machine is in place



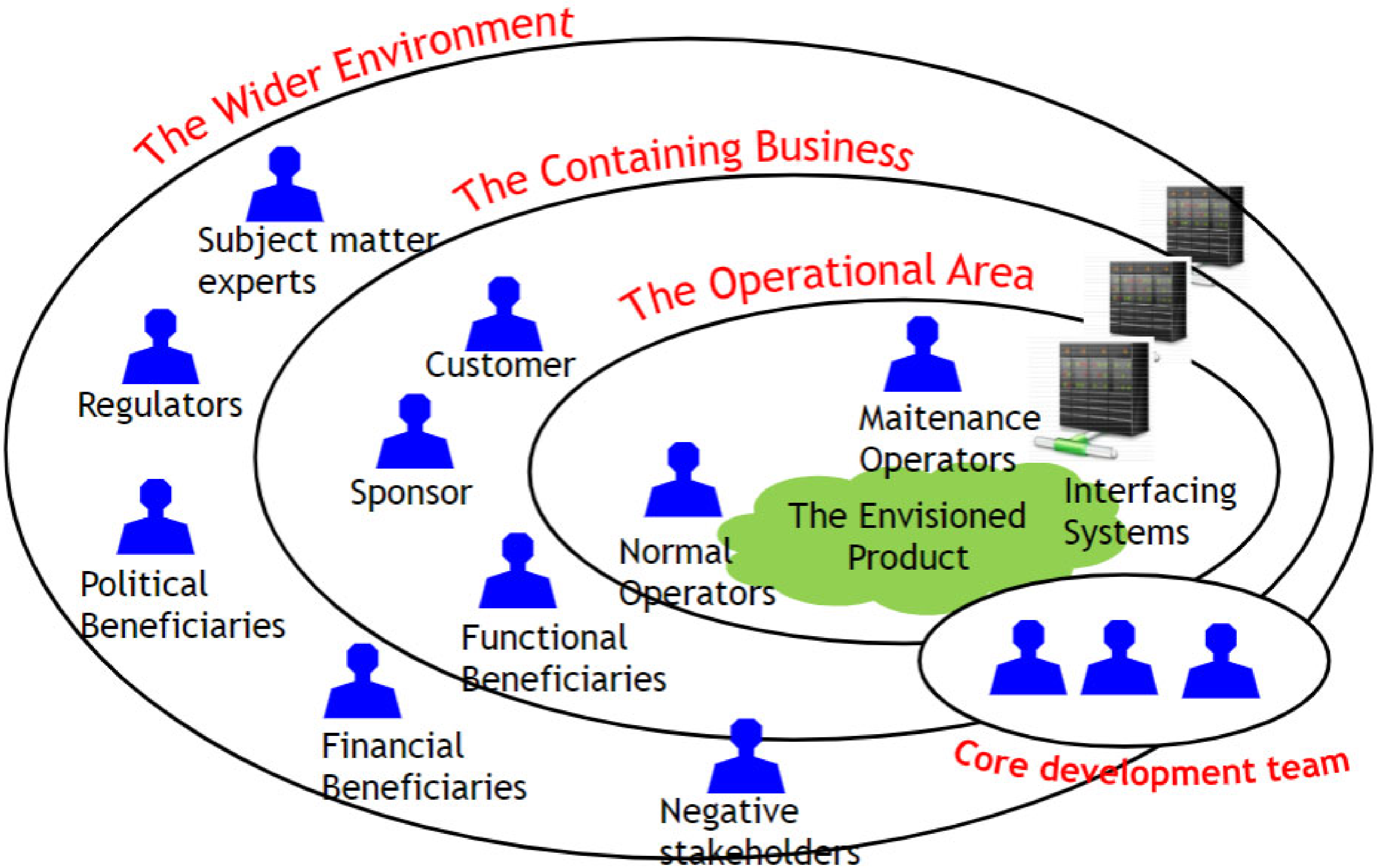
all the people and organisations affected by the project

part of the real world relevant to project

Stakeholders: who is affected by this project?

- The stakeholders are all the persons and organisations affected by the project
  - the project client and system users are stakeholders but there are many more!
- Beware of missing stakeholders: forgetting an important class of stakeholders can lead to missing important requirements

Stakeholders Onion Model



The operational area

All stakeholders who interact directly with the product

- The operators will operate the product
  - normal operator
  - IT operation, helpdesk
  - potentially many categories and profiles
- Adjacent systems
  - all external systems that have interfaces with the product
  - the stakeholders here are the people who have knowledge of these systems
- Core development team
  - project manager, architects, developers, validation team, ...

The containing business

All stakeholders who benefit from the product in some way

- The sponsor pays for the product development
  - Organisation management in customer-driven project
  - Sale/Marketing in market-driven project
  - Product manager, strategic program manager
- The customer buys the product once developed
- Functional beneficiaries are the people who benefit from the work done by operators in the operational area

The wider environment

All stakeholders who have an influence or interest in product

- Regulators and inspectors
  - legal requirements, safety inspectors, ...
- Financial and political beneficiaries
  - shareholders, politicians, ...
- Subject matter experts
  - domain expert, safety expert, usability expert, ...
- Negative stakeholders
  - do not want product, negatively affected by product, competitors
- Hostile stakeholders
  - attackers, criminals, terrorists, ...

Practical hints

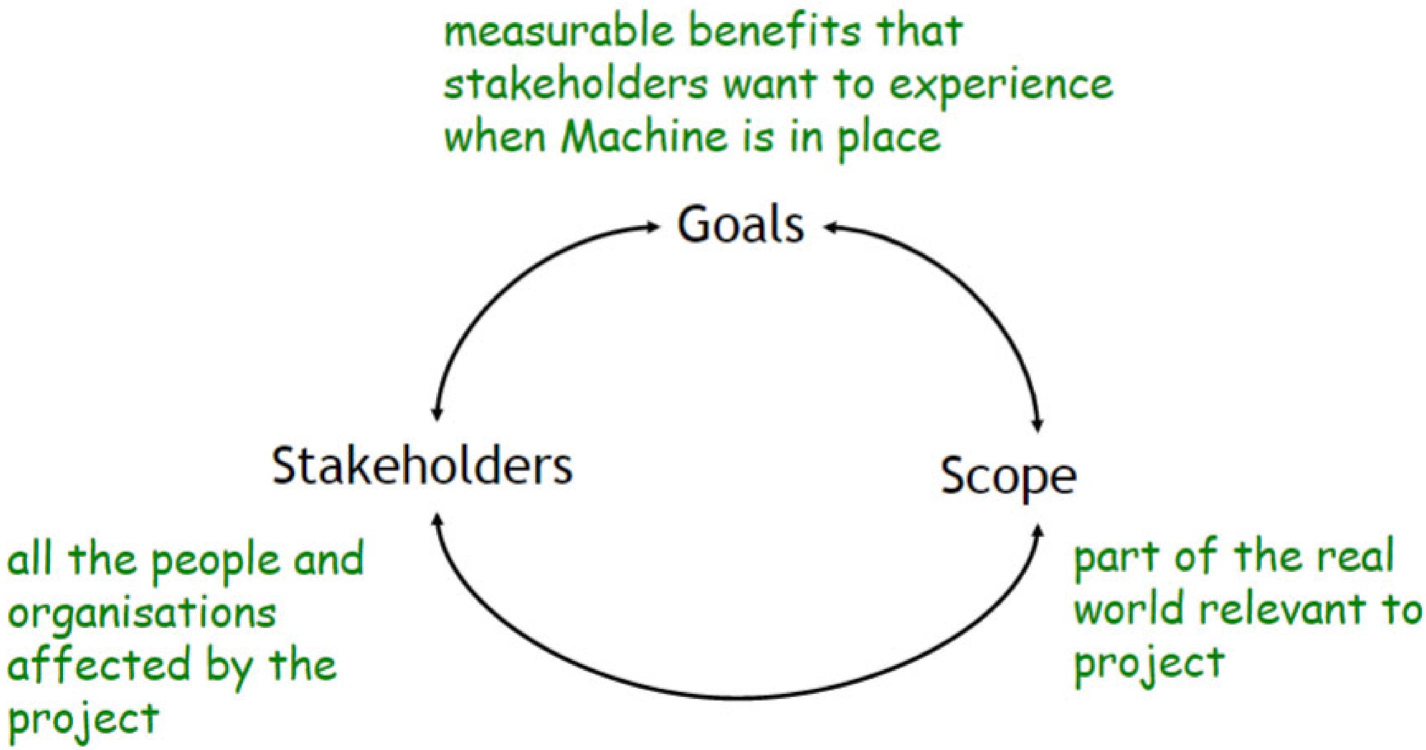
Use the stakeholders onion model as a checklist

Update your list of stakeholders during

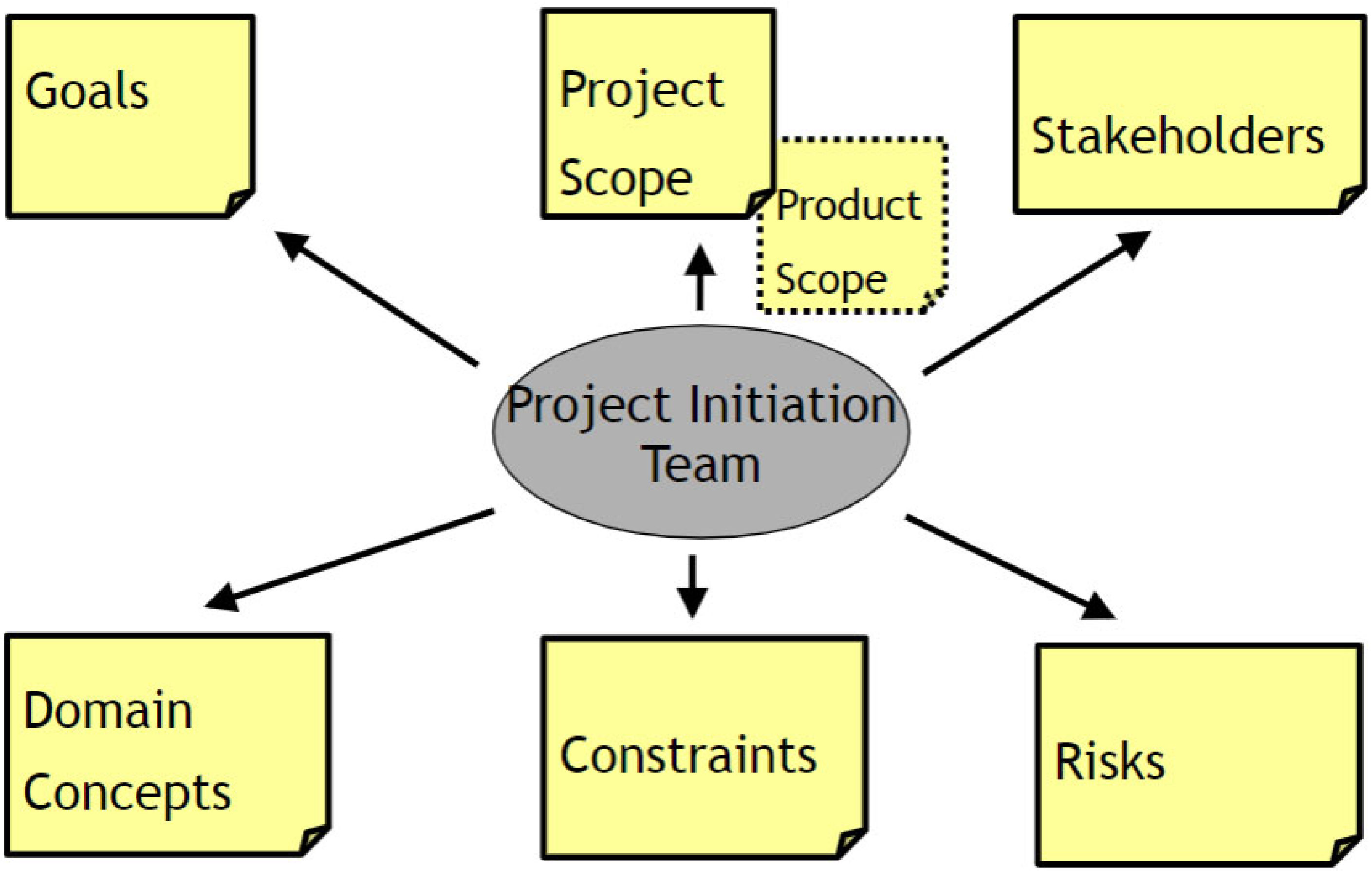
subsequent requirements elicitation and

throughout the project

Iterative Discovery



Other important project initiation deliverables



Domain concepts

- Start writing glossary of main domain concepts
- Clarifying terminology early on helps reducing misunderstandings!
- Can be modelled as Entity-Relationship diagram, i.e., a conceptual model

Constraints

- Project constraints
  - Time constraints
  - Financial constraints
- Solution constraints
  - Design and implementation constraints
  - Other systems with which your product must interact
  - Mandated (or forbidden) off-the-shelf and open source applications

Risks

- Identify main risks and actions to control them
  - consider process risks and application domain risks
  - identify risks by negating project goals

  Example for ambulance system

  Goal: Improve 14 minute response rate

  -> Risk: 14 minute response rate does not improve

  Goal: No reported incident left unresponded

  -> Risk: some reported incident left unresponded

Checklist: is the project worth doing?

- Clear Vision
  - are the goals clear and unambiguous?
  - are they measurable?
  - do they provide benefits to organisation?
- Feasibility
  - is it possible to achieve the goals within allotted time and budget?
  - are stakeholders willing to be involved?
- Risk Management
  - have risks been identified?
  - are there plans to track and control the risks?

Summary

- Good start is critical to project success
  - Stakeholders, Goals, Scope
  - Need measurable goals that truly matters
  - Start building project glossary
  - Identify constraints and risks
  - Feasibility analysis – cost/benefit, schedule
  - If too vague or not viable, stop project if you can
- Can take a few hours to several days or weeks